

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 28	
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Class	CSE A2
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Title of Assignment:	Develop a Java application demonstrating Remote Method Invocation (RMI) using stub and skeleton components for remote communication.
Practice questions **Output Screenshots Solve questions on given link and attach screen shot of leetcode with your login and o/p**	1. Create a remote calculator service using Java RMI 2. Create a Java application using RMI where the server converts temperature values between Celsius and Fahrenheit. The client sends input temperature and receives converted output. TemperatureService.java import java.rmi.Remote; import java.rmi.RemoteException; public interface TemperatureService extends Remote { double celsiusToFahrenheit(double celsius) throws RemoteException;

```
    double fahrenheitToCelsius(double fahrenheit)
            throws RemoteException;
}

TemperatureServiceImpl.java

import java.rmi.RemoteException;
import java.rmi.server.UnicastRemoteObject;

public class TemperatureServiceImpl
        extends UnicastRemoteObject
        implements TemperatureService {

    public TemperatureServiceImpl() throws RemoteException {
        super();
    }

    public double celsiusToFahrenheit(double celsius)
            throws RemoteException {
        return (celsius * 9 / 5) + 32;
    }

    public double fahrenheitToCelsius(double fahrenheit)
            throws RemoteException {
        return (fahrenheit - 32) * 5 / 9;
    }
}

TemperatureServer.java

import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;

public class TemperatureServer {
```

```
    public static void main(String[] args) {
        try {
            TemperatureService service =
                    new TemperatureServiceImpl();

            Registry registry = LocateRegistry.createRegistry(1099);

            registry.rebind("TemperatureService", service);

            System.out.println(
                "Temperature server is running...");
        } catch (Exception e) {
            System.out.println("Server error: " + e.getMessage());
            e.printStackTrace();
        }
    }
}
```

TemperatureClient.java

```
import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
import java.util.Scanner;

public class TemperatureClient {
    public static void main(String[] args) {
        try {
            Registry registry =
                    LocateRegistry.getRegistry("localhost", 1099);

            TemperatureService service =
                    (TemperatureService) registry.lookup(
                        "TemperatureService");
```

```
Scanner sc = new Scanner(System.in);

System.out.println("Temperature Conversion");
System.out.println("1. Celsius to Fahrenheit");
System.out.println("2. Fahrenheit to Celsius");
System.out.print("Enter choice: ");

int choice = sc.nextInt();

System.out.print("Enter temperature: ");
double temperature = sc.nextDouble();

if (choice == 1) {
   double result =
      service.celsiusToFahrenheit(temperature);

   System.out.printf(
      "%.2f Celsius = %.2f Fahrenheit%n",
      temperature, result);

} else if (choice == 2) {
   double result =
      service.fahrenheitToCelsius(temperature);

   System.out.printf(
      "%.2f Fahrenheit = %.2f Celsius%n",
      temperature, result);

} else {
   System.out.println("Invalid choice.");
}

sc.close();
```

```
        } catch (Exception e) {
            System.out.println("Client error: " + e.getMessage());
            e.printStackTrace();
        }
    }
}
```

Output Screenshots (if applicable)



Conclusion

Thus, a temperature conversion application was successfully developed using Java RMI. The application allows the client to send temperature values to the server for conversion between Celsius and Fahrenheit and receive the converted result. This practical helped in understanding remote method invocation, remote interfaces, RMI registry, and client-server communication in Java.